FIRST-DIFFERENCE LABELS AND A COUPLED-PROFILE REFINEMENT FOR A SENSITIVITY CONSTRUCTION

Abstract. The OpenAI family-132 construction of a superquadratic separation between ordinary sensitivity and block sensitivity uses a labelled regular tournament with no head-coherent set of sixteen vertices. We give an elementary directed adaptation of classical first-difference colouring that supplies the exact labels on an infinite parameter family, with polynomial logarithmic bit time per edge query and no witness advice. We also refine the construction's coupled sensitivity estimates: candidate target repairs usually cross two child thresholds, and target and gate candidate counts obey joint constraints. We prove the resulting six-branch ordinary and five-branch joint zero-side bounds. On one fixed explicit seed, exact arithmetic certifies an improvement of the amplified exponent bound from $2.015410903091582180\ldots$ to $2.015411547311478476\ldots$. The former already follows from immediate refinements of the published proof. The contribution is a deterministic instantiation and a local proof refinement, not a new separation architecture or a record-exponent claim.

1. Scope and attribution

For a total Boolean function $f : \{0,1\}^n \to \{0,1\}$, write x^D for the input obtained by flipping the coordinates in D. Its *sensitivity* $s(f,x)$ counts coordinates v with $f(x^{\{v\}}) \neq f(x)$. Its *block sensitivity* $\mathrm{bs}(f,x)$ is the largest number of pairwise disjoint nonempty sets D whose separate flips change $f(x)$. Put $s(f) = \max_x s(f,x)$ and $\mathrm{bs}(f) = \max_x \mathrm{bs}(f,x)$.

The pinned OpenAI manuscript [5] proves that $\mathrm{bs}(f)/s(f)^2$ is unbounded for total Boolean functions and obtains a fixed exponent greater than two by composition. Its construction adapts gated tournaments from Meiburg [4]; the latter's headline comparison is with spectral sensitivity λ, not ordinary sensitivity s. Since $\lambda \leq s$, a lower bound in terms of λ does not by itself imply the ordinary-sensitivity separation. OR balancing and composition are established tools; see Ambainis–Sun [1] and Tal [7].

This note proves two statements within the family-132 framework. First, a classical first-difference rule has a useful arbitrary-orientation head-coherence bound, yielding a succinct deterministic replacement for the source's probabilistic labelling. Second, a finer analysis of the source's candidate clauses gives stronger coupled profile inequalities. All Boolean predicates and the zero-block construction remain those of [5].

First-difference methods, including position-plus-own-bit local asymmetry, are classical: see Cole–Vishkin [2], Vishkin's notes [8], and the hypercube colouring in Gishboliner–Milojević–Sudakov–Wigderson [3]. The directed application below is an elementary adaptation, and historical novelty of its exact formulation is not assured. The numerical comparison later in the note is deliberately against an immediate refinement of the published proof on the *same seed*. An exponent exceeding 2.0154 already follows from that baseline.

2. Explicit labels for the exact tournament condition

Definition 2.1 (Head coherence)**.** A tournament directs exactly one edge between each pair of distinct vertices. Given labels $\ell(i,j)$ on its edges $i \to j$, a vertex set I is *head-coherent* if there are labels m_j, $j \in I$, such that

$$\ell(i,j) = m_j \qquad (i,j \in I,\ i \to j).$$

A vertex with no incoming edge in I imposes no restriction on m_j.

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The condition required in [5, Lemma 2.1] is a regular tournament on

$$k = 2M^2 + 1, \qquad r = \left\lceil \sqrt{M} \right\rceil, \qquad M \geq 3, \tag{1}$$

with outdegree M^2, labels in $\{0, \ldots, r-1\}$, and no head-coherent sixteen-set. Independent labels give the source's existence bound $k^{16} r^{-104} \leq 3^{16} M^{-20} < 1$.

Theorem 2.2 (First-difference bound)**.** *Let $h \geq 1$ and $q \geq 2$. Give any subset of $\{0, \ldots, q-1\}^h$ any tournament orientation. For $x \to y$, let u be the first position at which $x_u \neq y_u$, and set*

$$\ell(x, y) = (u-1)(q-1) + x_u - \mathbf{1}\{x_u > y_u\}. \tag{2}$$

These $h(q-1)$ labels have no head-coherent set of more than $h+2$ vertices.

Proof. At a fixed head digit b, the map $a \mapsto a - \mathbf{1}\{a > b\}$ is a bijection from the digits other than b to $\{0, \ldots, q-2\}$. Thus equal labels at a fixed head mean equal first-difference positions and equal source digits there. Positions have disjoint palettes.

Partition a coherent set by first digit into buckets. A vertex receiving an edge from its own bucket has an incoming label with position at least two. It cannot also receive an edge from outside that bucket, whose position would be one. It therefore sends an edge to every outside vertex. Every bucket of size at least two contains such a vertex. Two buckets of size at least two would consequently require both directions between a chosen vertex of each. Hence there is at most one large bucket B.

If B exists, choose $z \in B$ receiving an edge from B. Every vertex outside B receives an edge from z, and all outside buckets are singletons. If two outside vertices existed, their mutual edge would give its head two position-one labels with distinct source digits, one from B and one from the other singleton. Fixed-head injectivity excludes this. There is therefore at most one vertex outside B.

If all buckets are singletons, each vertex has indegree at most one; summing indegrees gives $|I|(|I|-1)/2 \leq |I|$, so $|I| \leq 3$. Inside a large bucket, delete its common first digit and subtract the palette offset. Coherence is preserved. The one-digit bound is three, and the preceding argument gives $C_h \leq \max\{3, C_{h-1} + 1\}$ for the maximum possible size. Induction yields $C_h \leq h + 2$. □

Remark 2.3. For $q \geq 3$ the bound is sharp over arbitrary orientations. Start with three words differing only in the last digit, oriented as a directed triangle. Successively add a word with a nonzero digit one position earlier than all preceding nonzero digits, directing all preceding vertices toward it. Each new head receives one label, giving $h+2$ coherent vertices. The proof also works for unequal radices q_u, using $\sum_u (q_u - 1)$ labels. If the last radix is two, the base bound is two and the resulting bound is $h+1$.

Theorem 2.4 (Exact infinite family)**.** *For every integer $q \geq 3$, let*

$$M = [13(q-1)]^2, \qquad r = 13(q-1), \qquad k = 2M^2 + 1. \tag{3}$$

There is a uniform deterministic tournament on k vertices, of outdegree M^2, with labels in $\{0, \ldots, r-1\}$ and no head-coherent sixteen-set. Each edge orientation and label can be evaluated using $O(\log M)$ storage bits and $O((\log M)^2)$ bit operations, with no precomputed witness or edge table.

Proof. On $\{0, \ldots, k-1\}$ use the cyclic orientation

$$i \to j \quad \Longleftrightarrow \quad 1 \leq (j - i) \bmod k \leq M^2. \tag{4}$$

There are exactly M^2 successors per vertex, and odd k ensures one direction per pair. Encode each index with thirteen base-q digits, including leading zeros, and use (2). This uses $13(q-1) = r = \left\lceil \sqrt{M} \right\rceil$ labels.

There are enough words: for $q = 3$, $k = 913953 < 1594323 = 3^{13}$. For $q \geq 4$,

$$q^{13} - k > q^4(q^9 - 2 \cdot 13^4) - 1 > 0,$$

since $4^9 = 262144 > 57122 = 2 \cdot 13^4$. Theorem 2.2 bounds coherent sets by fifteen. All integers involved have $O(\log M)$ bits. A fixed number of ordinary integer operations extracts thirteen digits and evaluates (2)–(4); schoolbook arithmetic gives the bit bound. The algorithm is uniform in q, and the family is unbounded. □

Remark 2.5 (Extent of the classical correspondence). Uncompressed labels (u, x_u) use the classical position-plus-source-digit idea. Compression in (2) preserves equality at each *fixed head*, which is precisely what Definition 2.1 uses. It need not preserve a global proper-colouring or walk-colouring property. The bucket argument proves the directed bound separately.

3. The inherited Boolean recursion

Fix $d \geq 1$, $M \geq 3$, and a labelled regular tournament with the parameters in (1) and no head-coherent sixteen-set. Here d is recursion depth; the digit length $h = 13$ is independent of it. Put

$$L = d + 2, \qquad H_j = d + 1 - j, \qquad N_j = LM(kr)^j \quad (0 \leq j \leq d). \tag{5}$$

The following predicates are exactly those of [5]. Partition the base coordinates into M disjoint L-sets E_v, let $z_v = \mathbf{1}_{E_v}$, and set

$$D(x) = \min_v \operatorname{dist}(x, z_v), \qquad P_{0,p}(x) = \mathbf{1}\{D(x) \leq d + 1 - p\}, \quad 1 \leq p \leq H_0. \tag{6}$$

At level $j \geq 1$, arrange the input into kr children $x_{i,t}$, where $0 \leq i < k$ and $0 \leq t < r$. Write $Q = H_{j-1}$. For $1 \leq p \leq H_j$, clause i for predicate index p requires

$$\begin{aligned} P_{j-1,Q}(x_{i,t}) &= 1 \quad (0 \leq t < r), \\ P_{j-1,Q-p}(x_{v,\ell(i,v)}) &= 0 \quad (i \to v). \end{aligned} \tag{7}$$

Let $P_{j,p}$ be the OR of all these clauses. The first line consists of *targets*, and the second of *gates*. Each condition in a fixed clause uses a distinct child.

Lemma 3.1 (Basic properties and zero blocks; inherited)**.** *These predicates are total and nested: $P_{j,1} \geq \cdots \geq P_{j,H_j}$. All reject zero. There are Mk^j pairwise disjoint nonempty blocks of size Lr^j whose separate flips at zero make every level-j predicate accept.*

Proof. All indices in (7) are valid, since $1 \leq Q - p < Q$. The formulas therefore define a value on every input. Nesting holds at the base, and increasing p decreases the gate index, strengthening the requirement that a nested predicate be zero. Targets are unchanged.

At the base, $D(0) = L > d$, and flipping E_v produces a centre, where all predicates accept. Inductively every target at zero is false. Given a preceding sensitive block, copy it into all r children of one row and take the union. Those targets accept after the flip, while every other row remains zero, so every outgoing gate passes. Distinct rows or preceding block indices give disjoint unions. Their number is $k \cdot Mk^{j-1}$ and their size is $r \cdot Lr^{j-1}$. □

Definition 3.2 (Ordinary and joint profiles)**.** For $b \in \{0, 1\}$, let

$$S_{j,b} = \max_{p,x:\, P_{j,p}(x)=b} \big|\{v : P_{j,p}(x^{\{v\}}) = 1 - b\}\big|.$$

The joint profile is

$$J_{j,b} = \max_{\substack{p<p',\, x: \\ P_{j,p}(x)=P_{j,p'}(x)=b}} \big|\{v : P_{j,p}(x^{\{v\}}) = P_{j,p'}(x^{\{v\}}) = 1 - b\}\big|.$$

Both predicates use the same input and the same raw coordinate. Empty maxima are zero; in particular $J_{d,0} = J_{d,1} = 0$. Always $0 \leq J_{j,b} \leq S_{j,b}$.

Lemma 3.3 (Base profiles; inherited)**.** *At level zero,*

$$S_{0,0} \le d+1, \qquad S_{0,1} \le LM, \qquad J_{0,0} = J_{0,1} = 0.$$

The displayed source recurrence uses the looser bound $S_{0,0} \le L$.

Proof. Distinct centres have distance $2L = 2d+4$, so at most one centre can lie within distance $d+1$ of a rejecting input. A successful flip must move toward that centre from distance $R+1$, where $R \le d$ is the predicate's radius. There are at most $R+1 \le d+1$ such coordinates. The one-side bound is the input dimension. Finally D is 1-Lipschitz on adjacent inputs, so one flip cannot cross two distinct integer radii. □

4. A coupled candidate refinement

At a fixed level write the preceding actual profiles as

$$a = S_{j-1,0}, \qquad b = S_{j-1,1}, \qquad c = J_{j-1,0}, \qquad e = J_{j-1,1}. \tag{8}$$

Thus $c \le a$ and $e \le b$. These letters are separate from the final seed sensitivity bound A and zero-block bound B.

Theorem 4.1 (Refined coupled profiles)**.** *For the predicates above and any admissible labels,*

$$S_{j,0} \le F_0(a,b,c,e), \qquad J_{j,0} \le F_J(a,b,c,e), \tag{9}$$

$$S_{j,1} \le M^2 a + rb, \qquad J_{j,1} \le M^2 c + rb, \tag{10}$$

where

$$F_0(a,b,c,e) = \max\left\{ \begin{array}{lll} a+14c, & a+13c+b, & 15c+e, \\ 13c+b+e, & 14c+2e, & 12c+3e \end{array} \right\}, \tag{11}$$

$$F_J(a,b,c,e) = \max\left\{ \begin{array}{ll} a+14c, & a+13c+e, \quad 15c+e, \\ 14c+2e, & 12c+3e \end{array} \right\}. \tag{12}$$

All branches have nonnegative coefficients. Consequently these bounds can be propagated from componentwise upper bounds, with optional definitional capping $J_{j,b} \le S_{j,b}$.

The one-side inequalities (10) are inherited from the source. The refinement concerns its treatment of candidate clauses at a rejecting input. We prove every branch, including the joint transition statement.

4.1. **Candidate structure.** Fix a parent index and a rejecting input. Suppress the child-level prefix, write Q for the strongest child index, and $g < Q$ for the gate index. Let T be the clauses with exactly one failed target and no failed gate. Let G be the clauses with exactly one failed gate and no failed target. Every successful coordinate repairs a clause in these fixed lists: one raw bit can affect at most one condition of a clause. Write m_t for the missing-target position of a row $t \in T$.

Lemma 4.2. *The set T is head-coherent and $|T| \le 15$. Every edge between G and T points from G to T. Moreover $|G| \le 3$; if $|G| = 3$, it is a directed cycle and all three failed gates are internal to G.*

Proof. For $s \to t$ inside T, the passing gate gives $P_g = 0$, hence $P_Q = 0$, at its selected child in row t. The unique such child is m_t, so $\ell(s,t) = m_t$. This proves coherence. An edge from T to G would fail its gate because all children of a G row have $P_Q = 1$, hence $P_g = 1$. Thus no such edge exists.

Every internal outgoing edge of a G vertex is a failed gate, so each has internal outdegree at most one. Counting edges yields $|G|(|G|-1)/2 \le |G|$, hence $|G| \le 3$. With three vertices the only possibility is a directed cycle, using each vertex's unique failure. □

Call a target candidate *ordinary* if its missing target has $P_g = 1$ and $P_Q = 0$, and *joint* otherwise. An ordinary target has at most a repair coordinates. A joint target starts with $P_g = P_Q = 0$ and a successful repair makes both one, so it has at most c repair coordinates. Call a gate candidate ordinary if its failed gate child has $P_Q = 0$, and joint otherwise. An ordinary gate costs at most b repairs. A joint gate starts with $P_g = P_Q = 1$ and repairing P_g makes both zero, so it costs at most e.

Lemma 4.3. *There is at most one ordinary target. If $|G| \geq 2$, every target is joint. At most one gate candidate is ordinary, and all gate candidates are joint when $|G| = 3$.*

Proof. An ordinary target has no incoming edge from T: such an edge would force $P_g = 0$ at its missing target. A tournament has at most one vertex of indegree zero. Also an ordinary target's entire row has $P_g = 1$. Every G vertex therefore uses its unique failed gate on that row. When $|G| \geq 2$, its internal edge already uses one such failure, a contradiction. Finally every failed gate internal to G is joint, because all targets of its head row pass. With two gate candidates only the sink could be ordinary; with three, every failure is internal by Lemma 4.2. □

4.2. **All six ordinary zero cases.** The following table gives upper bounds for the combined repair cost. The possible overlap of repair coordinates only makes the union smaller.

$\|G\|$	Condition	Repair bound
0	No gate candidate	$a + 14c$
1	An ordinary target or an ordinary gate exists	$a + 13c + b$
1	Neither ordinary type exists	$15c + e$
2	The sink's failed gate is ordinary	$13c + b + e$
2	Both failed gates are joint	$14c + 2e$
3	The gate candidates form a directed cycle	$12c + 3e$

For completeness, the cardinality argument for each row follows.

No gate candidate. There are at most fifteen targets and at most one is ordinary. Since $c \leq a$, their cost is bounded by $a + 14c$, including when no ordinary target exists or T is empty.

One gate candidate, an ordinary type present. Suppose first that the failed gate is ordinary. If it points into T, its child has $P_Q = 0$ and so is that row's missing target; its label is m_t. Every other edge from G into T passes and also has label m_t. Thus $T \cup G$ is head-coherent, whether or not the failure points into T, and $|T| \leq 14$.

Suppose instead that an ordinary target exists. Its entire row has $P_g = 1$, so the sole gate candidate's unique failure points into that row. The ordinary target is a source inside T and has just one incoming edge in $T \cup G$. Assign its head label to that edge's label; no internal incoming edge competes. All other heads retain their coherent labels. Again $T \cup G$ is coherent and $|T| \leq 14$. This reassignment of the exceptional head label is necessary; its missing-target position need not label the failed edge. The target cost is at most $a + 13c$, and the gate cost is at most b since $e \leq b$.

One gate candidate, neither ordinary type present. The general bound $|T| \leq 15$ gives target cost $15c$, and the gate costs e. The union need not be coherent in this case.

Two gate candidates, ordinary sink gate. The tail's failed gate is their internal edge; only the sink can fail outside G. If its ordinary failed gate points into T, its child has $P_Q = 0$, so its label is the head's missing position m_t. Every other edge into T passes and has its prescribed label. The internal edge of G imposes only one incoming label there, and there are no edges from T to G. Hence $T \cup G$ is coherent and $|T| \leq 13$. All targets are joint by Lemma 4.3; the gate cost is at most $b + e$.

Two gate candidates, both joint. Only the sink can have a failure outside G. All edges from G into T therefore have the expected head label except possibly at the head of that one failure. Delete that target if it exists. The remaining union is coherent, so $|T| - 1 + 2 \leq 15$ and $|T| \leq 14$. All target repairs are joint, giving $14c + 2e$.

Three gate candidates. All failures are internal to their directed cycle, whose heads each have one incoming edge. All edges into T pass. Thus $T \cup G$ is coherent and $|T| \le 12$. All candidates are joint, giving $12c + 3e$.

These cases exhaust the possibilities. Taking their maximum and then maximizing over rejecting inputs proves (11).

4.3. **All joint zero transitions and the one side.** Fix parent indices $p < p'$ and an input where both predicates are zero. Their gate indices obey $g' = Q - p' < g = Q - p < Q$. Form T at the stronger parent index p'. Form G there too, but retain only candidates whose unique failed gate also fails at the weaker index p.

A coordinate making both parents accept satisfies some clause at p'. Before the flip this clause has exactly one failed condition. A target failure puts it in T. If it is a gate failure, every other gate already passes at p', hence also at p. All targets pass as well. If that same gate did not fail at p, the weaker parent would already accept, a contradiction. Thus this clause lies in the retained G. The fixed lists cover every simultaneous success even when the witness clause depends on the flipped coordinate.

The previous candidate arguments apply to these lists at gate index g'. Deleting gate candidates changes none of their premises. Target repairs have the same costs a, c. Every retained gate, including a gate ordinary relative to P_Q, now costs at most e: its distinct predicates $P_{g'}$ and P_g start at one, and satisfying the stronger gate makes both zero. Replace b by e in the six rows. The fourth row $13c + 2e$ is dominated by the fifth, $14c + 2e$. The remaining five rows are exactly (12). If there are no distinct parent indices the joint profile is zero by definition.

For ordinary one-to-zero transitions, fix one initially satisfied clause. Every successful rejection breaks it, by one of its r targets or M^2 gates. These cost at most rb and $M^2 a$, respectively. For joint rejection at indices $p < p'$, fix a clause satisfied at p'; it also satisfies p. Since both parents reject after the flip, this clause must fail at the weaker index p. A target failure costs at most b. A gate failure changes P_g from zero to one; initially $P_{g'} = 0$, and nesting forces $P_{g'} = 1$ afterward, so it costs at most c. This proves (10) and completes the proof of Theorem 4.1.

Remark 4.4 (What the refinement does and does not assert). The displayed source bounds are

$$S_{j,0} \le 16a + b + 3e, \qquad J_{j,0} \le 16a + 3e, \tag{13}$$

with the same one-side bounds. The new argument uses joint target repairs and coupled counts; it is not just replacement of 16 by 15. For example, when $c = e = 0$, it gives $S_{j,0} \le a + b$ and $J_{j,0} \le a$.

One cannot simply assume $|T| + |G| \le 15$. On a transitive set of fifteen target rows, give every row the two child states $(0, 2)$, where states $0, 1, 2$ denote $(P_g, P_Q) = (0, 0), (1, 0), (1, 1)$. Add a gate row $(2, 2)$ pointing to all targets. Give all target edges label zero and all gate edges label zero except its edge to the last target, labelled one. There are fifteen target candidates and one gate candidate; their full sixteen-set is incoherent at that last head. Likewise, three gate candidates can form a directed cycle. These examples delimit the local counting argument; they are not asserted to be full regular recursive seeds or to attain all global profile maxima simultaneously.

5. Propagation, balancing, and a fixed-seed comparison

5.1. **The inherited OR and composition principles.**

Proposition 5.1 (Seed and amplification)**.** *Suppose s_0, s_1 bound $S_{d,0}, S_{d,1}$. For an integer $C \ge 1$, the OR of C disjoint copies of $P_{d,1}$ is a total nonconstant function f, rejects zero, and has*

$$n = CLM(kr)^d, \qquad s(f) \le A = \max\{Cs_0, s_1\}, \qquad \mathrm{bs}(f, 0) \ge B = CMk^d. \tag{14}$$

If $B > A^2 > 1$, the m-fold disjoint iterate F_m has dimension n^m and satisfies, for $\alpha = \log B / \log A > 2$,

$$F_m(0) = 0, \quad s(F_m) \le A^m, \quad \mathrm{bs}(F_m, 0) \ge B^m \ge s(F_m)^\alpha. \tag{15}$$

In particular $\mathrm{bs}(F_m, 0)$ *tends to infinity.*

Proof. At a zero of the OR every copy rejects, so sensitivities sum to at most Cs_0. At a one, two accepting copies prevent any single-bit rejection; with exactly one accepting copy, the sensitivity is at most s_1. Lemma 3.1 supplies CMk^d disjoint zero-sensitive blocks and proves nonconstancy.

For total f, g, a bit of child i is sensitive for $f \circ g$ exactly when that bit is sensitive for g and outer coordinate i is sensitive for f at the vector of child outputs. Hence $s(f \circ g) \le s(f)s(g)$. If $g(0) = 0$, disjoint zero-sensitive families $U_1, \dots, U_a$ for f and $V_1, \dots, V_b$ for g give the nonempty, pairwise disjoint blocks $U_i \times V_j$ for $f \circ g$. Flipping one changes exactly the outer coordinates in U_i. Thus $\mathrm{bs}(f \circ g, 0) \ge \mathrm{bs}(f, 0)\,\mathrm{bs}(g, 0)$. Iteration proves (15), and $B > 1$ gives growth. □

Proposition 5.2 (Published coarse consequence with explicit labels)**.** *For each* $d \ge 1$*, choose*

$$q(d) = \max\left\{3, 1 + \left\lceil \frac{3^{d+1}}{13} \right\rceil\right\}, \qquad M = [13(q(d) - 1)]^2. \tag{16}$$

Then the explicit seed with $C = M$ *satisfies*

$$n = (d+2)M^2(kr)^d, \qquad A = 2(d+2)M^{d+1}, \qquad B = M^2k^d, \tag{17}$$

and $B/A^2 \ge 2^d/[4(d+2)^2]$*. At* $d = 9$ *this is at least* $128/121 > 1$*, so Proposition 5.1 gives a uniform power separation. These quantitative estimates are inherited from* [5]*; explicit labels are supplied by Theorem 2.4.*

Proof. Equation (16) gives $M \ge 9^{d+1}$. Normalize the published profiles by M^{j+b} and let u_j, v_j be the maxima of the normalized ordinary and joint profiles. Put $\varepsilon = \max\{r, 16\}/M \le 2/3^{d+1}$. Equations (10) and (13) give

$$u_j \le (1+\varepsilon)u_{j-1} + 3v_{j-1}, \qquad v_j \le \varepsilon u_{j-1} + 3v_{j-1},$$

with $u_0 \le L, v_0 = 0$. The corresponding equality comparison sequences U_j, V_j satisfy simultaneously $U_j \le 2L$ and $V_j \le \varepsilon L(3^j - 1)$ for $j \le d$. Indeed the bound for V_j follows from the preceding bounds, and summing the increments for U gives

$$U_j \le L + \varepsilon L\left[\frac{3}{2}(3^j - 1) - j\right] < 2L,$$

since $\varepsilon \le 2/3^{d+1}$. Thus $S_{j,b} \le 2LM^{j+b}$. Substituting in (14) with $C = M$ gives (17), and $k/M^2 > 2$ gives the ratio. □

5.2. **The source-derived baseline used here.** Before comparing numerical consequences of Theorem 4.1, we include the following immediate improvements of the source proof in the baseline. Use $S_{0,0} \le d + 1$ rather than $d + 2$, and $|T| \le 15$ rather than 16, still charging every target repair to a. One or two gate candidates cost at most $b + e$; three cost $3e$. For simultaneous acceptance all gates cost e. Thus the immediate source-derived zero-side updates are

$$S_{j,0} \le 15a + \max\{b + e, 3e\}, \qquad J_{j,0} \le 15a + 3e. \tag{18}$$

Together with (10), these are valid for every $M \ge 3$ with admissible labels. The large-M restriction in Proposition 5.2 is only sufficient for its convenient scalar estimate, not a restriction on exact recurrence iteration.

The OR-copy count can also be optimized without a new combinatorial theorem. If $W = Mk^d > s_0$, the function

$$\frac{\log(CW)}{\log\max\{Cs_0, s_1\}}$$

increases up to $C = s_1/s_0$ and decreases afterward, where its denominator is positive. Below that point the denominator is constant; above it the derivative has the sign of $\log(s_0/W) < 0$.

The two positive integers bracketing s_1/s_0 therefore suffice. Both immediate refinements and OR optimization belong to the baseline.

5.3. **Certified comparison on one unchanged seed.** The following finite result compares proved upper bounds on sensitivity, not exact sensitivity values. It changes neither the labels nor the Boolean function between the two analyses.

Theorem 5.3 (Fixed explicit seed)**.** *Set*

$$\begin{aligned} q = 1001, \quad d = 17, \quad M = 169000000, \quad r = 13000, \\ k = 57122000000000001, \qquad C = 164900179. \end{aligned} \tag{19}$$

Use Theorem 2.4's labels and the recursion above. The seed dimension and zero-block bound are exactly

$$n = C \cdot 19M(kr)^{17}, \qquad B = CMk^{17}. \tag{20}$$

The integer n has 373 decimal digits. Iteration of the immediate baseline gives a bound A_{old}; iteration of Theorem 4.1 gives a strictly smaller bound A_{new}. For $\alpha_* = \log B / \log A_*$, certified enclosures are

$$2.015410903091582180 < \alpha_{\mathrm{old}} < 2.015410903091582181, \tag{21}$$

$$2.015411547311478476 < \alpha_{\mathrm{new}} < 2.015411547311478477. \tag{22}$$

Composition of this one explicit seed therefore satisfies (15) with $\alpha = \alpha_{\mathrm{new}} > 2.0154$. The strict improvement is from A_{old} to A_{new} at fixed B, n; the inequality $\alpha > 2.0154$ itself already holds for the baseline.

Proof and arithmetic certificate. The exact definitions of the two integers are short recurrences. Start with $v_0 = (18, 19M, 0, 0)$. For $v_j = (a, b, c, e)$, set

$$v_{j+1} = (z,\ M^2a + rb,\ \min\{w, z\},\ \min\{M^2c + rb, M^2a + rb\}), \tag{23}$$

where (z, w) is either the right side of (18) or (F_0, F_J). Iterate exactly seventeen times. If (a_*, b_*) are the first two final entries, define $A_* = \max\{Ca_*, b_*\}$. At the terminal level the actual joint profiles are zero; their final comparison entries are not used. All operations are on integers and all cappings are valid by Definition 3.2.

Evaluation of these recurrences gives the exact rational enclosures

$$\begin{aligned} 3187070 \cdot 10^{143} < A_{\mathrm{old}} < 3187071 \cdot 10^{143}, \\ 3186720 \cdot 10^{143} < A_{\mathrm{new}} < 3186721 \cdot 10^{143}. \end{aligned}$$

In particular $1 < A_{\mathrm{new}} < A_{\mathrm{old}}$. For both values A_*, exact integer powers verify

$$A_*^{20154} < B^{10000} \le A_*^{20155}. \tag{24}$$

Hence both exponent bounds exceed 2.0154, and the common positive B with $A_{\mathrm{new}} < A_{\mathrm{old}}$ proves a strict improvement.

For the finer enclosures, the accompanying certificate evaluates $\log A_*, \log B$ with correctly rounded 70-digit decimal logarithms. The adjacent representable values enclose each exact logarithm. Dividing the lower numerator endpoint by the upper positive denominator endpoint with downward rounding, and conversely with upward rounding, encloses their ratio. The resulting intervals are contained strictly in (21)–(22). The numerical operations use the documented logarithm and rounding contracts of Python's Decimal implementation [6]. The companion JSON records all integers, the longer intervals, and the exact-power checks (24). Equation (23) and parameters (19) independently specify the certificate. Finally Theorem 2.4 supplies the labels, and Proposition 5.1 proves the stated Boolean-function bounds. □

No numerical search is needed to define this seed. The chosen C optimizes the immediate baseline's two bracketing integer candidates using certified interval comparison; it is then kept unchanged in the new analysis. There is no assertion of a globally optimal exponent, a minimal dimension, or simultaneous attainability of all profile bounds.

Both calculations use only the no-head-coherent-sixteen-set contract. For this particular seed, $k < 1001^6$, so the thirteen-digit vertex words share seven leading zeros. Deleting them and subtracting the common label offset, Theorem 2.2 actually bounds coherent sets by eight. This is a direct consequence of the label theorem. Neither calculation exploits that stronger label-specific bound. The comparison therefore isolates the coupled-profile refinement at a fixed contract; it is not an optimized analysis even of this explicit seed.

6. Uniformity, verification, and limitations

The seed is described by (d, q, C) and a fixed program, requiring $O(\log d + \log q + \log C)$ parameter bits. There is no label advice. For (16) with $C = M$, one can give only d and derive the other integers: $\log M = O(d)$ and $\log n = O(d^2)$. The parameter description then has $O(\log d)$ bits, while deriving its large integers takes time polynomial in d, not necessarily in $\log d$.

Polynomial logarithmic edge access is distinct from evaluation of the whole n-bit function. A bottom-up evaluator computes every child's predicate vector once. There are $O(n)$ nodes and at most $dk(r + M^2) = O(dk^2)$ condition tests per internal node. Base distances can be obtained by scanning each leaf once, since

$$D(x) = \mathrm{wt}(x) + L - 2 \max_v \mathrm{wt}(x|_{E_v}).$$

A conservative bound is $O(ndk^2(\log n)^2)$ bit operations and $O(n(d + 1))$ bits for input, predicate values, and bookkeeping. Thus expansion of an evaluation circuit has polynomial cost in the input size; neither the description nor the generator requires a 2^n-entry truth table.

The companion files in `topics/sensitivity-separation/` of the repository include the generator, local profile diagnostics, exact recurrence implementation, and `checks/seed-certificate.json`. The bounded generator check exhausts 129088 small tournaments and also checks a sharp coherent fifteen-set and parameter arithmetic. The profile checker exhausts small nested-state models and checks explicit candidate-count templates. These are finite regression checks; the infinite label theorem and all-input profile theorem rest on the proofs above. The fixed-seed comparison is additionally an exact arithmetic calculation. A single reproduction command from that topic directory is

```
python3 scripts/reproduce.py -fetch-sources.
```

It verifies pinned source hashes and runs the bounded checks serially; it materializes no large tournament or Boolean truth table.

The scope of the contribution is limited. The first-difference technique, the gated recursion, OR balancing, and composition are inherited methods. The label theorem supplies an elementary directed adaptation with exact uniform parameters. The profile theorem refines a local step of the published proof. No priority guarantee, record exponent, or optimality claim is made.

AI assistance and verification status

The arguments, code, and draft were developed with OpenAI Codex using multiple GPT-6 Astra agents for construction, adversarial checking, and writing. Separate agent reviews checked the saved mathematical arguments and code, and the companion reproduction run passed. This describes AI-assisted verification, not human peer review. The results of this note have not been formalized in Lean, and no Lean kernel replay is claimed. The written proofs and identified arithmetic artifacts state the evidence supporting the conclusions.

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